Start Here

Session 1 · Advanced SQL for Modern Data Engineering

Welcome to the Voltkart challenge

This pack is a hands-on SQL challenge built around Voltkart, an online electronics and appliances retailer. You'll answer real analytical questions and build change-driven load logic — the kind of work a data engineer does every day — using T-SQL on SQL Server.

What's in this pack

File	What it's for
00 · Start Here	This overview
01 · Problem Statement	The business background and what you're being asked to do
02 · Assignment Questions	The 11 questions to solve (10 + 1 bonus)
03 · Importing the Dataset	Step-by-step load of the data into SQL Server
04 · Data Dictionary	Every table and column explained
05 · Hints	A nudge per question — tables and concepts, never the answer
data/	The eight CSV files and a `load.sql` helper

How to work through it

1. Read the **Problem Statement** so the business context is clear.
2. Follow **Importing the Dataset** to load the data into SQL Server.
3. Work through the **Assignment Questions** in order — they build on each other.
4. Reach for the **Hints** only after you've tried a question yourself.

What to submit

1. Put **all your solution files** — the `.sql` for every question and a short

results write-up — in a **single public GitHub repository**, and submit the **repository link**. That link is your submission.

2. For the optimization question, include in the repo a

screenshot of the execution plan and the `SET STATISTICS IO` output before and after your fix.

3. *(Optional, good to have)* Add a short **video** — a Loom or any screen recording — walking through your solution and explaining your decisions. It's excellent practice for the stakeholder communication data engineers do every day, and it strengthens your submission.

How your work is graded

Axis	What we look for
Correctness	Your outputs and the acceptance criteria are met against the data
Logic & approach	You pick the right tool — window vs self-join, MERGE vs separate DML, SARGable predicates
Readability	CTEs over deep nesting, clear naming, comments, consistent formatting
Communication	Your write-up or Loom explains your decisions to a stakeholder

> **Tip** AI tools like Copilot, Claude, and Cursor are allowed and encouraged. What we assess is your understanding — be ready to explain every query, especially the recursive CTEs, the MERGE logic, and the execution-plan fix.